						3
						4
						5
						6
			30 (100%)			المجموع

29

Part (1): Choose the correct answer for the following questions and transfer it to the table below:

1	2	3	4	5	6	7	8
D	C	A	A	C	A	B	A

1. Which hazard pictogram indicates a material can catch fire easily?

A) Toxic B) Corrosive C) Radioactive D) Flammable

2. chemist needs to add a very precise volume during a titration. Which apparatus should be used?

A) Test tube B) Beaker C) Burette D) Funnel

3. 3. An analytical balance has an accuracy of ..... g?

A) ± 0.0001 B) ± 0.001 C) ± 0.01 D) ± 1.0

4. According to the rule "like dissolves like", which compound is expected to be most soluble in water?

A) NaCl B) C_6H_6 C) Hexane D) $CaCO_3$

5. Metal has a mass of 0.015 kg and a volume of 5.0 mL. Its density is:
A) 0.75 g/mL B) 10 g/mL (C) 3.0 g/mL D) 0.033 g/mL

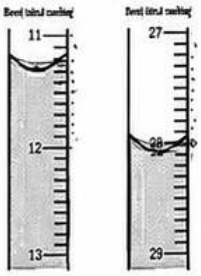
6. Which intermolecular force usually produces the lowest boiling point?
(A) London dispersion forces B) Hydrogen bonding C) dipole forces D) None

7. Which apparatus is used to heat the hydrated salt strongly?
A) water bath (B) Crucible C) Burette D) Pipette

8. A hydrate loses 0.90 g of water when heated. If the original hydrate mass was 2.00 g, the percentage of water is:
(A) 45.0% B) 22.5% C) 55.0% D) 90.0%

Part (2): Fill in the Blank

11.3 28.1



1. Based on the burette readings shown in the figure, calculate the volume delivered (ΔV)
28.1 − 11.3 = 16.8 mL (1)

2. The volume of an irregular solid can be determined using the density / graduated method.

3. If part of the solid dissolves during density determination, the calculated density will be ...higher.... than the true value. (1)

4. The boiling point generally ....increases.... as atmospheric pressure increases. (1)

5. Concerning hydrated salts Experiment, the mass of water lost equals the mass of the ..hydrated.... salt minus the mass of the anhydrous... salt. (1)

6. Using the experimental data below, determine the value of X and write the formula of the hydrate. Show calculations.

$MgSO_4 \cdot XH_2O \rightarrow MgSO_4 + XH_2O$

Measurement	Value
Mass of hydrate sample	4.0
Mass of $MgSO_4$ after heating	2.1 g
Molar mass of H_2O	18.0 g/mol
Molar mass of $MgSO_4$	120.4 g/mol

$m_{H_2O} = 4 - 2.1 = 1.9\,g$ $\quad m_{MgSO_4} = 2.1\,g$

$n_{H_2O} = \frac{1.9}{18} = 0.1056\ mol$ $\quad M_{MgSO_4} = 120.4\ g/mol$

$X = \frac{\text{moles } H_2O}{\text{moles } MgSO_4} = \frac{0.1056}{0.01745\ mol} = \boxed{X = 6}$ (3)

$\boxed{MgSO_4 \cdot 6H_2O}$



7. Based on the apparatus shown in the figure below, identify the experiment ..Dumas method.... (1)
.....and name the liquid present inside the flask. ..Volatile liquid.... (1)

8. Determine the Molecular Weight and density of an Unknown, depend on data on the table.

$M = \frac{mRT}{PV}$ (3)

$d = \frac{m}{V} = \frac{8.55 \times 10^{-4}}{300} = \boxed{2.85 \times 10^{-6}\ g/mL}$ ✓

$N = \frac{8.55 \times 10^{-4} \times 0.08206 \times 372.5}{0.98 \times 0.3}$

$N = 0.088\ g/mol$ ✓

Parameter	Value
Mass of empty flask + foil + rubber band	0.082615 g
Mass of flask + condensed vapor	0.083470 g
Volume of flask	300.0 mL
Water bath temperature	99.5°C
Pressure	748 torr
R	0.08206 L·atm·mol^{-1}·K^{-1}